What Authenticity Costs in Metres: A Certified Budget for Cryptography and Sensing in UAV Navigation

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***Abstract*—A recent line of work certifies that an unmanned aerial vehicle (UAV) completes its mission against an adversary who stays just below an intrusion detector's threshold. The certificate bounds the position error as the product of three factors: the delay the adversary can hide, the rate γ at which delayed navigation input becomes position error, and a dynamics factor. We observe that authentication moves the first two factors in opposite directions. Authenticating a navigation source lowers γ, because a forged fix is rejected rather than fused. Signing, transmitting and verifying the authenticator raises the delay. The certificate is therefore a budget in which cryptography and sensing trade, and it decides the sign of the trade in metres.**

We make this budget explicit and measure both sides. On the sensing side, 674 post-onset samples from real global positioning system (GPS) spoofing and jamming flights show that a spoofed estimator degrades up to $1.74\times$ faster than one fed degraded but honest fixes. Within the first half-second of staleness, however, the two cannot be told apart. On the cryptographic side, a response-time analysis of a controller area network (CAN) bus, using a base load derived from a real UAVCAN capture, shows that the binding cost of post-quantum signatures (ML-DSA and SLH-DSA, standardised by the US National Institute of Standards and Technology, NIST) is bus occupancy, not latency. ML-DSA-44 costs 119.0 ms per message, yet classic CAN admits only 16.4 signed messages per second. SLH-DSA cannot be carried at a usable rate at all.

Four results follow. (i) Under the certificate's supremum discipline, every per-message signature scheme that fits on the bus pays across the certified window. Authentication also rescues the tightest corridor, which the unauthenticated certificate cannot cover. (ii) Matched to the horizon the detector actually admits, the data cannot show any benefit at the parent's operating point. The two error envelopes differ by 1 cm, well inside measurement noise, and authentication pays only from $\theta_\times \approx 0.259$–$0.284$ s. A sound certificate is therefore not necessarily a good design oracle. (iii) Deployed freshness windows of 30–165 s never bind against detector-scale delays. Where the signing clock is advanced from GNSS time, as MAVLink 2 prescribes at GPS lock, a spoofer who pushes that clock forward converts the delay adversary into a denial adversary. (iv) An attacker who floods invalid Ed25519 signatures saturates the verifier using 0.105 of the bus. Large post-quantum signatures are immune on classic CAN, but not on CAN with flexible data rate (CAN FD). A selection rule returns the cheapest certified configuration. Every number in the paper regenerates from released data, and a defined set of simulator and hardware experiments replaces each remaining modelled input.

***Index Terms*—UAV security, certified robustness, post-quantum cryptography, broadcast authentication, CAN bus, GNSS spoofing, real-time analysis**

I. Introduction

An autonomous unmanned aerial vehicle (UAV) navigates on inputs it does not produce itself. Satellite fixes from a global navigation satellite system (GNSS) receiver, position reports relayed by peers in a swarm, and commands from a ground station all arrive over links an adversary can reach. Two families of defence address this exposure, and they rarely meet. Cryptographic authentication proves where a message came from. Navigation-message authentication for GNSS [1], [2], message signing on the Micro Air Vehicle Link (MAVLink) protocol [3], [4], and authentication schemes for the controller area network (CAN) bus [5]–[7] all belong here. Certified robustness proves that a navigation stack tolerates a bounded perturbation of its inputs. A recent composition [8] joined an intrusion detector to such a certificate. For an adversary who delays relayed messages by just less than the detector's threshold θ, it bounds the worst-case deviation from the planned route as

$$\rho(\theta) \;=\; \Delta(\theta)\cdot\gamma\cdot e^{LT_c} \;\le\; m, \tag{1}$$

where $\Delta(\theta)$ is the delay the adversary can hide, γ the rate (in metres per second) at which stale input becomes position error, L a Lipschitz constant of the navigation dynamics, T_c a re-anchoring horizon, and m the lateral margin of the safe corridor.

The observation: Two of the four factors in Eq. (1) are fixed by physics (L) and by regulation (m). The other two are security-engineering choices, and authentication moves them in opposite directions. Authenticating a navigation source lowers γ, because an estimator that rejects a forged fix stops absorbing the forger's offset. Authenticating a message raises Δ, because signing, transmitting and verifying the authenticator take time, and stale input is stale regardless of why it is stale. Authentication is therefore both a credit and a debit against one certified budget. Neither the cryptographic nor the certified-robustness literature can say which effect wins, because the two sides are measured on different instruments: a cycle counter on a microcontroller and a flight log of a spoofed receiver.

This paper: We write the budget down, derive when authentication increases the certified margin, and measure both sides on real data and real specifications. The results do not all point the same way, and the disagreement is itself a finding.

- **The crossover and its two readings** (Section III, Proposition 1, Theorem 1). Under the parent certificate's discipline of one supremum rate per configuration, authentication pays whenever its cost Δ_{auth} is below $\Delta(\theta)(\gamma_{\text{u}}/\gamma_{\text{a}}-1)$. With the measured rates, every per-message signature scheme that fits on the bus meets this across the certified window. It also rescues the 2 m corridor that the unauthenticated certificate cannot cover: $\rho = 1.88$ m with ML-DSA-44, against 2.65 m without. When the rates are matched to the horizon the detector actually admits, the picture changes at tight settings. Within half a second of staleness, the spoofed and the honest-but-degraded estimators have accrued the same error to within 1 cm, so the data cannot show a benefit. On the measured envelopes, authentication pays from $\theta_{\times} = 0.259$ s (a group message authentication code, MAC) to 0.284 s (ML-DSA-44).
- **The cost is a cliff, not a slope** (Proposition 3). We consider the post-quantum cryptography (PQC) signatures that NIST standardised as Federal Information Processing Standards (FIPS): the module-lattice-based digital signature algorithm ML-DSA (FIPS 204) [9] and the stateless hash-based SLH-DSA (FIPS 205) [10]. We use a base load derived from a real capture of UAVCAN (now DroneCAN, the CAN protocol linking a flight controller to its peripherals). On that bus, the worst-case transmission latency of a signed message is flat up to an admissible rate $r^{\star}$ and unbounded beyond it. On classic CAN, $r^{\star}$ is 16.4/s for ML-DSA-44 and 5.0/s for SLH-DSA-128s. A discrete-event simulation of 254,992 transfers stays within the analytical bound.
- **Freshness does not bound delay, and the clock can turn delay into denial** (Proposition 5). Deployed freshness windows exceed the largest delay the detector admits ($Hs = 10$ s) by a factor of 3–17: 60 s for MAVLink 2, and 30–165 s for Galileo's Open Service Navigation Message Authentication (OSNMA). Where the signing clock is advanced from GNSS time, a spoofer who pushes it past the window makes every legitimate signed packet stale.
- **Induced verification** (Proposition 6). Forged signatures convert the defender's verification cost into delay. The classical Edwards-curve signature Ed25519 is the scheme most exposed: 0.105 of the bus suffices to saturate a verifier that has a 20 % share of the central processing unit (CPU). ML-DSA-44 is immune on classic CAN because its signature is too expensive to transmit, but not on CAN FD. A slot-budget mitigation bounds the induced delay to one verification.
- **A certified selection rule** (Algorithm 1). For each detector setting, corridor, transport, adversary class and post-quantum requirement, the rule returns the cheapest configuration (scheme, batch size, mitigation) that keeps the setting certified, or reports that none exists.

Scope and honesty: The sensing side rests on the same three real flights of one airframe in hover as the parent [11]. The authenticated rate is a *proxy*: it is measured on a jamming flight whose receiver kept a degraded three-dimensional fix, not on an estimator that rejected GNSS outright (Section IV). The cryptographic compute side rests on published Cortex-M cycle counts scaled to the flight controller's clock. The transport side rests on standard frame models and a base load derived from frame rates measured on a real bus. Each number carries a provenance tag, and Section IV-G specifies the simulator and hardware experiments that replace each modelled input. Because every number in this paper is generated from result files, a replacement propagates to the text without hand editing.

II. Preliminaries and Problem Statement

A. The Parent Certificate

We inherit the system model and constants of the parent manuscript [8], summarised here so that the paper is self-contained.

Navigation loop: A UAV running the PX4 autopilot [12] fuses GNSS fixes, measurements from an inertial measurement unit (IMU), and barometric height in an extended Kalman filter (EKF). It also receives relayed corrections from peers over a multi-hop mesh. When GNSS is gated out, the EKF dead-reckons on the IMU, and its position error grows.

Residual delay: A multi-hop forwarding detector D_θ flags any hop whose forwarding delay exceeds a per-hop tolerance θ. An adversary controlling H relays on a route can hide at most

$$\Delta(\theta) \;\le\; H\,\min(\theta, s) \tag{2}$$

seconds of delay, where s is the contact-window slack. A delay beyond s misses its window and is flagged at any threshold. The parent measures $s = 5$ s (cap 4.84 s over 15,392 hops). It also measures a benign residual ceiling of 0.178 s, below which θ cannot go without false alarms, and adopts $H = 2$ and the operating point $\theta = 0.25$ s.

Interface rate: The rate γ is estimated with the baseline-corrected estimator

$$\gamma_\delta(\tau) \;=\; \frac{e(t_0+\tau) - e(t_0)}{\tau}, \tag{3}$$

where e is the horizontal position error of the EKF output, t_0 the attack onset, and τ the staleness. The parent evaluates its certificate at the supremum over all 674 post-onset samples, $\gamma = 1.625$ m/s.

Tube: If the navigation drift is L-Lipschitz and the filter re-anchors at least once every T_c, Grönwall's inequality [13] bounds the deviation by Eq. (1), with $L = 1.1814$, $T_c = 1$ s, and hence $E \triangleq e^{LT_c} = 3.259$. The corridor family $m \in \{2, 5, 10, 20\}$ m brackets the contingency volumes of the Joint Authorities for Rulemaking of Unmanned Systems (JARUS) Specific Operations Risk Assessment (SORA) [14]. At $\theta = 0.25$ s the unauthenticated tube is $\rho = 2.65$ m. The setting is therefore certified for $m \ge 5$ m but not for $m = 2$ m. We call the set of θ for which $\rho(\theta) \le m$ the *certified window*. Its lower edge is the benign floor of 0.178 s.

TABLE I
ADVERSARIES AND WHAT AUTHENTICATION BUYS AGAINST EACH.

Adversary	Effect of authentication
$\mathscr{A}_{\text{spoof}}$: forges GNSS or peer position data	Forged data fail verification and are rejected. The estimator stops following the forger, so γ falls from γ_{u} to γ_{a}.
$\mathscr{A}_{\text{delay}}$: compromised relay forwards valid messages late	None, unless the signature covers a timestamp and the receiver enforces a freshness window τ_{fresh}; then the hidden delay is capped at $\tau_{\text{fresh}} + \varepsilon_{\text{clk}}$.
$\mathscr{A}_{\text{ins}}$: insider holding a group key	Group message authentication codes (MACs) give no protection, since the insider can MAC anything. Only asymmetric signatures or delayed key disclosure, as in Timed Efficient Stream Loss-tolerant Authentication (TESLA), authenticate the source.
$\mathscr{A}_{\text{deny}}$: selectively withholds unauthenticated sources beyond T_c	Re-anchoring fails. An authenticated source that is still delivered within T_c is *necessary* for any certified floor (Proposition 4). A jammer that also denies the authenticated source is outside this model.
$\mathscr{A}_{\text{iv}}$: injects transfers with invalid signatures	Each costs t_{ver} before rejection, so the defence's cost becomes delay (Proposition 6).
$\mathscr{A}_{\text{time}}$: shifts the receiver's GNSS-disciplined clock	Erodes or inverts the freshness check (Proposition 5).

B. Threat Model

We retain the parent's two adversaries and add three. Table I summarises what authentication does against each.

Three points in Table I are easy to get wrong.

First, a validly signed message forwarded four seconds late is still validly signed. Authentication bounds delay only through a timestamp and a freshness check, and that check is only as good as the receiver's clock.

Second, the MAVLink 2 signature is a 48-bit truncation of the Secure Hash Algorithm SHA-256 over a shared secret, the packet and a timestamp, not a public-key signature [3]. CAN-layer schemes such as vatiCAN [5], LeiA [6] and the CAN XL security layer (CANsec) [15] are likewise group-keyed. Against a compromised relay, which holds the key, they authenticate group membership, not origin. This has been noted for in-vehicle multicast [16].

Third, Galileo's Open Service Navigation Message Authentication (OSNMA) [1] authenticates the navigation data, not the ranging signal. Receivers must keep their clock within a loose time-synchronisation bound T_L (30 s for the authentication-data-and-key-delay types ADKD0 and ADKD4). A meaconer that replays the signal with a delay inside T_L therefore passes the check [17], [18]. For OSNMA, therefore, γ_{a} against a meaconer equals γ_{u}. Our measured γ_{a} applies to forgery that a verifier can detect.

Assumptions: The adversary knows θ, the scheme and the transport. Legitimate authenticated navigation messages are released periodically at rate r (10 Hz unless stated). The adversary's injected CAN frames do not out-prioritise the legitimate stream: a higher-priority flood is ordinary bus flooding, which identifier-based intrusion detection on the UAVCAN capture of the Hacking and Countermeasure Research Lab (HCRL) already handles [8], [19].

TABLE II
NOTATION.

θ, H, s	detector tolerance (s), attacker hops, contact-window slack (5 s)
$\Delta(\theta)$	hidden delay $\le H\min(\theta, s)$ (s)
Δ_{auth}	authentication delay: $t_{\text{sign}} + R_{\text{tx}} + t_{\text{ver}} + t_{\text{proto}}$ (s)
Δ_{eff}	effective delay entering the tube, Eq. (5)
γ_{u}, γ_{a}	delay-to-position rate, unauthenticated / authenticated (m/s)
$\delta_{\text{u}}(\tau)$, $\delta_{\text{a}}(\tau)$	error accrued within staleness τ (m)
L, T_c, E	Lipschitz constant, re-anchoring horizon, $E = e^{LT_c}$
m, ρ	corridor margin, tube radius (m)
$C(\theta)$, $B(\theta)$	benefit threshold, affordability budget (s)
$\theta_{\times}$	smallest θ at which authentication pays (s)
σ, n, C_{f}	tag bytes, frames per tag, worst-case frame time
U_0, r, $r^{\star}$	base bus load, signed rate, admissible rate
τ_{fresh}, ε_{clk}, ν	freshness window, clock error, clock push rate
ρ_{iv}, β, a	verification amplification, CPU share, attacker bus share
k	batch size (messages per signature)

C. Problem Statement

Let κ be an authentication configuration: a scheme, a batch size k (one signature per k messages) and a flooding mitigation. Let $\mathscr{T}$ be the transport. The *certified budget* is

$$\rho(\theta,\kappa) \;=\; \Delta_{\text{eff}}(\theta,\kappa)\cdot\gamma_{\text{eff}}(\kappa)\cdot e^{LT_c} \;\le\; m, \tag{4}$$

$$\Delta_{\text{eff}}(\theta,\kappa,t) \;=\; \min\big(\Delta(\theta),\, \tau_{\text{fresh}} + \varepsilon_{\text{clk}}(t)\big) + \Delta_{\text{auth}}(\kappa,\mathscr{T}), \tag{5}$$

with $\gamma_{\text{eff}} = \gamma_{\text{u}}$ for unauthenticated and $\gamma_{\text{eff}} = \gamma_{\text{a}}$ for authenticated sources. The design problem is

$$\begin{aligned} \min_{\kappa\in\mathscr{K}} \quad & \text{cost}(\kappa) \\ \text{s.t.} \quad & \rho(\theta,\kappa) \le m, \quad \kappa \text{ admissible on } \mathscr{T}, \\ & \kappa \text{ secure against } \mathscr{A}, \end{aligned} \tag{6}$$

where admissibility requires bus and processor utilisation below one, and security requires source authentication against $\mathscr{A}_{\text{ins}}$, post-quantum security where mandated [20], and resistance to $\mathscr{A}_{\text{iv}}$. We ask five research questions (RQs):

RQ1 How much does authentication lower γ, and over which staleness horizon?

RQ2 What does authentication cost in delay on flight-controller-class hardware, and is the binding term latency or occupancy?

RQ3 Where is the crossover in $(\theta, \text{scheme}, \text{transport})$?

RQ4 Does a freshness window bound the hidden delay under GNSS-coupled clocks?

RQ5 How much delay can an attacker induce through verification, and how much does mitigation recover?

Table II collects the notation. Fig. 1 shows how the answers are assembled.

III. METHODOLOGY I: THE CERTIFIED BUDGET

The framework (Fig. 1) has five stages: inputs with provenance, component models, the budget, decisions, and a validation loop. This section develops stage (3). Section IV develops stages (2) and (5). Proofs omitted here are in Appendix A.

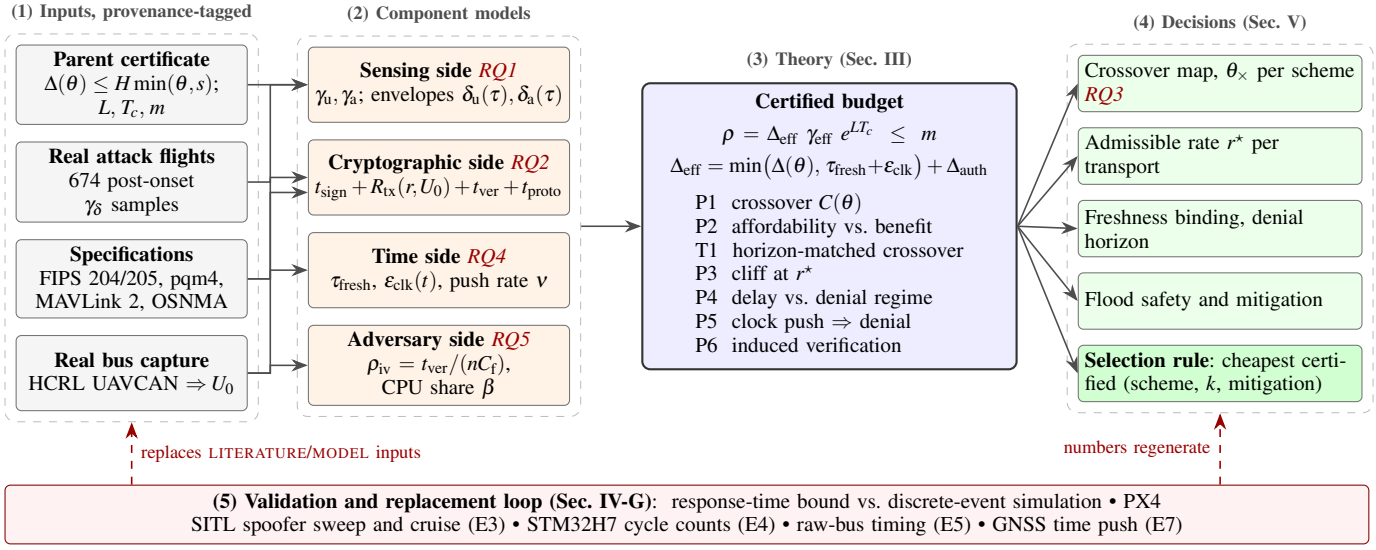


Fig. 1. Methodological framework. (1) Every input carries a provenance tag: measured on real flights or buses, derived, taken from a specification, or modelled. (2) Four component models quantify the two sides of the trade, sensing and cryptography, together with time and the flooding adversary. (3) The certified budget composes them into one inequality in metres, from which the propositions follow. (4) The decisions an engineer needs are read off the budget. (5) Simulation and hardware experiments replace modelled inputs with measured ones; because every number in the paper is generated from result files, a replacement propagates to the text automatically.

A. The Crossover under the Supremum Discipline

Write $E = e^{LT_c}$. In the delay regime, where re-anchoring holds by construction and the freshness cap does not bind, $\Delta_{\text{eff}} = \Delta(\theta) + \Delta_{\text{auth}}$.

Proposition 1 (Crossover)**.** *Let authenticating the navigation path change the rate from γ_u to $\gamma_a < \gamma_u$ at cost Δ_{auth}. Authentication strictly reduces the certified error if and only if*

$$\Delta_{\text{auth}} < C(\theta) \triangleq \Delta(\theta)\left(\frac{\gamma_u}{\gamma_a} - 1\right). \tag{7}$$

Equivalently, for $\theta < s$, authentication pays if and only if $\theta > \theta_\times \triangleq \Delta_{\text{auth}}/\big(H(\gamma_u/\gamma_a - 1)\big)$.

Proposition 2 (More affordable, less beneficial)**.** *Let $B(\theta) = m/(\gamma_a E) - \Delta(\theta)$ be the largest Δ_{auth} that keeps the authenticated configuration certified. For $\theta < s$, B decreases with slope $-H$ and C increases with slope $H(\gamma_u/\gamma_a - 1)$. Tightening the detector therefore leaves more room to afford authentication and less reason to want it.*

The two halves answer different questions, *can I pay for it?* and *is it worth paying for?*. An engineer who conflates them will over-invest in cryptography on a platform whose detector is already tight.

B. The Crossover with a Time-Varying Rate

Proposition 1 inherits the parent's choice of one supremum rate per configuration. That choice is sound for certification, but it can mislead design. The supremum of γ_u may be attained at a staleness the detector never admits, and the unauthenticated configuration is then charged for an error it cannot suffer. We therefore also state the crossover in terms of the measured error envelope.

Definition 1 (Error envelope)**.** For a source in state $x \in \{u, a\}$, let $\delta_x(\tau) = \sup_{\tau' \le \tau}\big[e(t_0 + \tau') - e(t_0)\big]$, taken over the observed onsets. By construction, δ_x is non-decreasing.

Theorem 1 (Horizon-matched crossover)**.** *Under the hypotheses of the parent certificate, with $\gamma\Delta$ replaced by the envelope $\delta(\Delta)$, authentication strictly reduces the certified tube radius if and only if*

$$\delta_a\big(\Delta(\theta) + \Delta_{\text{auth}}\big) < \delta_u\big(\Delta(\theta)\big). \tag{8}$$

Corollary 1 (Indistinguishability horizon)**.** *If $\delta_u(\tau) \le \delta_a(\tau)$ for all $\tau \le \tau_s$, then authentication cannot pay at any θ with $\Delta(\theta) \le \tau_s$, whatever its cost.*

Proof. For $\Delta(\theta) \le \tau_s$, monotonicity gives $\delta_a(\Delta + \Delta_{\text{auth}}) \ge \delta_a(\Delta) \ge \delta_u(\Delta)$, which contradicts Eq. (8). □

The corollary has a physical reading. A spoofer that walks the fix away gradually, as spoofers must to stay inside the EKF's innovation gate, needs time to out-run dead reckoning. Until it does, rejecting its data buys nothing.

C. The Cost Is a Cliff, Not a Slope

Decompose

$$\Delta_{\text{auth}} = \underbrace{t_{\text{sign}} + t_{\text{ver}}}_{\text{compute}} + \underbrace{R_{\text{tx}}}_{\text{transmission}} + \underbrace{t_{\text{proto}}}_{\text{protocol}}, \tag{9}$$

where R_{tx} is the worst-case response time of the transfer that carries a σ-byte authenticator in $n(\sigma)$ frames of worst-case duration C_f.

Proposition 3 (Cliff)**.** *Let signed transfers be released periodically at rate r on a fixed-priority CAN bus with periodic*

background traffic of utilisation U_0. Then $R_{\mathrm{tx}}(r)$ is finite for $r < r^\star$ and infinite for $r \geq r^\star$, where

$$r^\star \;=\; \frac{1-U_0}{n(\sigma)C_{\mathrm{f}}}. \tag{10}$$

Below $r^\star$, R_{tx} is bounded by the busy-period recurrence of Davis et al. [21]. Ordering schemes by σ captures only the transmission term: schemes that minimise it, such as TESLA, pay instead in t_{proto} through their key-disclosure delay.

The proposition is a direct consequence of level-i busy-period analysis: the busy period closes if and only if the total utilisation is below one. Its content for this paper lies in the magnitudes (Section V-B). Per-message latency is milliseconds, against budgets of hundreds of milliseconds, so the binding constraint is $r^\star$, not the latency.

D. Two Regimes, and Freshness Turned Against Itself

Proposition 4 (Two regimes)**.** *Under $\mathscr{A}_{\mathrm{delay}}$, re-anchoring holds by construction and the crossover decides. Under $\mathscr{A}_{\mathrm{deny}}$, which withholds every unauthenticated source for longer than T_c, the unauthenticated certificate is vacuous. An authenticated source that is still delivered at least once per T_c is then necessary for any certified floor, and, with the authenticated rate γ_{a}, sufficient for the tube bound.*

Proposition 5 (Freshness under a GNSS-coupled clock)**.** *(a) A freshness window reduces the hidden delay only if $\tau_{\mathrm{fresh}} + \varepsilon_{\mathrm{clk}}(t) < \Delta(\theta)$. (b) Suppose the verifier's signing clock is updated as the maximum of its own value and GNSS time, and rejects packets more than τ_{fresh} behind it. MAVLink 2 prescribes this update at GPS lock for systems without a real-time clock [3]; whether an autopilot also applies it continuously is implementation-specific, and experiment E7 tests it. Then an adversary who advances GNSS time at rate ν makes every legitimate signed packet fail freshness after $\tau_{\mathrm{fresh}}/\nu$ seconds. It thereby converts $\mathscr{A}_{\mathrm{delay}}$ into $\mathscr{A}_{\mathrm{deny}}$. If the update happens only at lock, a spoofer present at lock achieves the same with a single step.*

Part (b) is the coupling between time and attack that the drone-authentication literature has observed experimentally [22] and the GNSS-authentication literature analysed for OSNMA [17], [18], [23]. Here it is placed inside the certificate. The forward-only clock update prevents an attacker from rolling the clock back to admit replays, but it hands the attacker the opposite move.

E. Induced Verification

Proposition 6 (Verification amplification)**.** *Let a verifier with processor share β spend t_{ver} on each received transfer, and let an attacker occupy a fraction a of the bus with n-frame transfers carrying invalid signatures. Define the amplification ratio*

$$\rho_{\mathrm{iv}} \;=\; \frac{t_{\mathrm{ver}}}{nC_{\mathrm{f}}}. \tag{11}$$

The verification queue is stable if and only if $a\rho_{\mathrm{iv}} + r t_{\mathrm{ver}} < \beta$. Beyond this, the induced delay is unbounded and the certificate is vacuous. If verification of messages claiming a given source is capped at k_s per expected slot, and $k_s t_{\mathrm{ver}}/\beta < 1/r$, the induced delay is at most $(k_s - 1)t_{\mathrm{ver}}/\beta$ for any a.

Algorithm 1 Certified selection rule

Require: θ, m, transport $\mathscr{T}$, adversary class, PQ flag, regime
$\mathscr{F} \leftarrow \emptyset$
if regime = delay **and** $\Delta(\theta)\gamma_{\mathrm{u}}E \leq m$ **then**
  $\mathscr{F} \leftarrow \{(\mathrm{none}, 0)\}$ ▷ unauthenticated is certified
end if
for scheme q allowed by adversary class and PQ flag **do**
  for $k \in \{1, 2, 4, \ldots, 32\}$ **do**
    if $U(k) < 1$ **and** signer/verifier load $< \beta$ **then**
      if $(\Delta(\theta) + \Delta^{\mathrm{cert}}_{\mathrm{auth}}(q,k))\,\gamma_{\mathrm{a}}E \leq m$ **then**
        for mitigation $\mu \in \{\mathrm{none}, \mathrm{slot}\}$ **do**
          if flood-safe$(q, \mathscr{T}, \mu)$ by Prop. 6 **then**
            $\mathscr{F} \leftarrow \mathscr{F} \cup \{((q,k,\mu), \mathrm{cost})\}$
          end if
        end for
      end if
    end if
  end for
end for
return $\arg\min_{\mathscr{F}}$ cost, or INFEASIBLE if $\mathscr{F} = \emptyset$

ρ_{iv} is CPU time spent per unit of attacker bus time. It is small when a signature is expensive to transmit relative to verifying it, which is exactly the regime of large post-quantum signatures on slow buses.

F. Amortisation and the Selection Rule

Signing a Merkle root over k messages trades occupancy for latency:

$$\Delta_{\mathrm{auth}}(k) = \tfrac{k-1}{r} + t_{\mathrm{sign}} + R_{\mathrm{tx}} + t_{\mathrm{ver}}, \tag{12}$$

$$U(k) = U_0 + \tfrac{r}{k} n(\sigma) C_{\mathrm{f}} + r n(h\lceil \log_2 k \rceil) C_{\mathrm{f}}, \tag{13}$$

with h-byte hashes on the authentication path. A MAC-plus-periodic-signature hybrid has the same latency form with k equal to the signature period. Algorithm 1 solves Eq. (6) by exact enumeration of a finite space. In it, $\Delta^{\mathrm{cert}}_{\mathrm{auth}}$ denotes Δ_{auth} with the signing time replaced by its $1 - 10^{-6}$ quantile (Section IV).

IV. Methodology II: Measurement and Simulation Design

A. Inputs and Provenance

Table III lists every input with its provenance tag. Four tags are used: MEASURED (from a real capture in the released artifact), DERIVED (arithmetic on measured values), LITERATURE (a cited specification or benchmark), MODEL (a stated engineering model), and ASSUMPTION (a parameter we choose).

B. Sensing Side (RQ1)

Proxy for the authenticated rate: The rate γ_{a} should describe an estimator that does not follow an adversary. Two realisations satisfy this. In the first, a failed verification gates GNSS out and the EKF dead-reckons. In the second, GNSS is degraded but honest, and the EKF keeps fusing it. The

TABLE III
INPUTS AND PROVENANCE.

Input	Source	Tag
γ_u	178 post-onset samples, spoofing flight [11]	MEASURED
γ_a	496 post-onset samples, jamming flight (proxy for rejection)	MEASURED PROXY
$\Delta(\theta)$, L	parent artifact [8]	MEASURED
T_c, $\theta = 0.25$ s, m	parent choices	ASSUMPTION
U_0	HCRL frame rate and payload mix (measured, E5) × assumed 1 Mbit/s [19]	DERIVED
cycle counts	pqm4 [24], SLOTHY-M7 [25], [26], [27]	LITERATURE
tag sizes	FIPS 204/205 [9], [10], MAVLink [3]	LITERATURE
τ_{fresh}	MAVLink 2 (60 s), OSNMA (30–165 s) [23]	LITERATURE
frame timing	CAN/CAN FD frame model [21], [28], DroneCAN tail byte	MODEL
oscillator drift	XO/TCXO/OCXO classes	MODEL

released flights contain only the second. Throughout the post-onset samples, the jamming flight's receiver keeps a three-dimensional fix with 12–14 satellites; jamming raised its noise without denying the fix. We therefore take γ_u from the spoofing flight and γ_a from the jamming flight, both on the EKF signal the controller acts on, and call the latter a *measured proxy*.

The proxy is exact for an adversary-free estimator that keeps fusing honest GNSS. It is *not* a measurement of dead reckoning after rejection. Pure dead reckoning may be slower, since the noisy fixes are gone, or faster, since nothing bounds the inertial drift. Experiments E1 and E3 measure it directly (Section IV-G), and every conclusion that depends on γ_a is stated as conditional on the proxy.

We report three scenarios. The *headline* scenario uses the EKF signal on both sides, giving $\gamma_u/\gamma_a = 1.74$. The *conservative* scenario uses the largest jamming rate over either error signal, giving 1.32. The *per-flight secant* scenario gives 2.63.

Supremum ratios are not amenable to the bootstrap. We therefore also report the ratio of 95th percentiles. Its interval combines the two moving-block bootstrap intervals (block of ten samples, respecting within-flight autocorrelation) at their extremes, and is therefore conservative.

Envelopes and dead reckoning: For Theorem 1 we compute δ_u and δ_a as running maxima of the accrued error, with linear interpolation between samples. Each flight contributes a single onset, so the envelopes are empirical estimates rather than suprema over many onsets. We compare their differences with the residual noise of the error signal. To characterise the proxy we also fit $e(\tau) = v_0\tau + \frac{1}{2}b\tau^2$ to the jamming flight for $\tau < 5$ s. This is the error-growth form of Groves [29], with a velocity error v_0 and a residual acceleration bias b.

C. Cryptographic Side (RQ2)

Compute: Cycle counts come from pqm4 (Cortex-M4) [24] for FN-DSA-512 (the fast-Fourier-transform-over-NTRU-lattice signature, Falcon) and SLH-DSA-SHA2-128s/128f; from the SLOTHY Cortex-M7 implementation for ML-DSA-44 [25]; from Fujii and Aranha for Ed25519 [26]; and from cifra for the SHA-256 MAC [27]. We scale them to the 480 MHz STM32H7 microcontroller (MCU) of a Pixhawk 6 assuming equal cycles per operation. This is the input that hardware experiment E4 replaces.

ML-DSA signing is a rejection-sampling loop. We model the iteration count as geometric with mean 4.25 [9] and use its $1-10^{-6}$ quantile, a factor of $12.2\times$ the mean, as the certified signing time. This is a probabilistic bound: at 10 Hz it is exceeded about once every 28 hours.

Transport: For DroneCAN on classic CAN at 1 Mbit/s, a transfer carries seven payload bytes per eight-byte frame (one tail byte) plus a two-byte transfer cyclic redundancy check (CRC) [30]. An extended frame lasts at most

$$C_f = \left(54 + 8\cdot 8 + 13 + \lfloor (54 + 8\cdot 8 - 1)/4 \rfloor\right)\tau_{\text{bit}} = 160\,\mu\text{s}$$

including worst-case bit stuffing [21]. For CAN FD we use 64-byte frames with a 1/5 Mbit/s bit-rate switch and the field sizes of ISO 11898-1 [28]. For MAVLink 2 over serial we use 8N1 framing (eight data bits, no parity, one stop bit), so that each byte occupies ten bit times. We derive the base load from the benign segments of the HCRL UAVCAN capture [19] (508–619 frames/s). Experiment E5 decoded the raw capture: across its 10 scenarios the mean payload is 6.96–7.33 bytes per frame, not the 8-byte worst case, and 0.86–0.96 of transfers are multi-frame. With the measured payload mix, $U_0 \in [0.076, 0.095]$, and we use the maximum. The frame rate and payload mix are measured; the bitrate is an assumption, so U_0 is tagged DERIVED.

R_{tx} is the busy-period bound of a lowest-priority signed transfer below 24 periodic background streams carrying U_0. We check it against a discrete-event simulation (DES) of frame-level arbitration with random background phases. The simulation shares the periodic background model, so it checks the analysis for self-consistency and tightness; it does not validate the model. The HCRL capture cannot do so either: its software timestamps place consecutive 8-byte frames closer than one frame time, implying 1.44–2.11 Mbit/s on a bus whose maximum is 1 Mbit/s, so they are not frame-accurate. Validating bus timing needs E5's hardware arm.

Protocol: TESLA [31], [32] authenticates a message only when its key is disclosed, one interval later. We sweep the interval over $\{0.05, 0.1, 0.5, 1\}$ s and add a 10 ms synchronisation bound.

D. Time Side (RQ4)

We model $\varepsilon_{\text{clk}}(t) = |y_0|t + \frac{1}{2}Dt^2$ for a crystal oscillator (XO, ±20 ppm), a temperature-compensated one (TCXO, ±2 ppm) and an oven-controlled one (OCXO, ±0.05 ppm). We evaluate Proposition 5(a) for the deployed windows and Proposition 5(b) for push rates ν from 1 ms/s to a step.

E. Adversary Side (RQ5)

The DES models the bus at frame granularity. Legitimate transfers win arbitration frame by frame and wait for at most one attacker frame. Attacker transfers (Poisson arrivals) use

the remaining capacity. A first-in-first-out verification queue is served at share $\beta = 0.2$. We compare three mitigations:

- a 4-byte group-key pre-filter, which stops an outsider but not an insider;
- a slot budget that verifies at most $k_s = 2$ candidates per expected legitimate slot;
- no mitigation.

Each configuration runs for four seeds of 60 s.

F. Selection Grid

Algorithm 1 is run over two γ scenarios, two regimes, two adversary classes, the post-quantum flag, two CAN transports, $\theta \in \{0.178, 0.25, 0.5, 1\}$ s and the four corridors, giving 512 cells. The cost is the added bus utilisation plus the processor load, with a small charge for slot budgeting.

G. Validation and Replacement Plan

Stage (5) of Fig. 1 consists of eight experiments, specified in the artifact's runbook:

E1 inertial γ from synthetic GNSS outages on the benign flight's IMU log;
E2 visual–inertial odometry (VIO) γ on EuRoC [33];
E3 a software-in-the-loop (SITL) PX4 sweep of spoofer walk rate, with an authenticated-rejection arm, in hover and in cruise at 5–15 m/s;
E4 STM32H7 cycle counts, including ML-DSA tails;
E5 payload-length mix from the HCRL capture (done, above) and bus timing on hardware;
E6 a hardware flood test;
E7 a GNSS time push against MAVLink 2 signing in SITL;
E8 an OSNMA time-offset replay.

Each experiment writes a result file that replaces a LITERATURE or MODEL input in Table III. E3 matters most, because it maps how the crossover of Theorem 1 moves with the spoofer's walk rate and with airspeed.

V. Results

A. RQ1: What Authentication Buys in γ

Fig. 2 shows the two rates as functions of staleness. Over all samples, the supremum ratio is $\gamma_u/\gamma_a = 1.625/0.933 = 1.74$. The 95th-percentile ratio is 1.95, with a conservative block-bootstrap interval of $[1.68, 2.67]$.

The structure matters more than the number. In the first second of staleness, both estimators accrue error at about 0.9 m/s, and the curves cannot be told apart. From about one second onwards the spoofer walks the solution away, while the jammed estimator's rate falls to 0.6–0.9 m/s and then decays. The fit to the jamming flight gives $v_0 = 0.857$ m/s and $b = -0.014$ m/s^2, with a root-mean-square error (RMSE) of 0.16 m over 45 samples. The growth is linear, which is consistent with a velocity bias in a degraded but still-fused estimator. Because this flight never lost its fix, the fit characterises the proxy and does not show what pure dead reckoning would do. That is the question E1 answers.

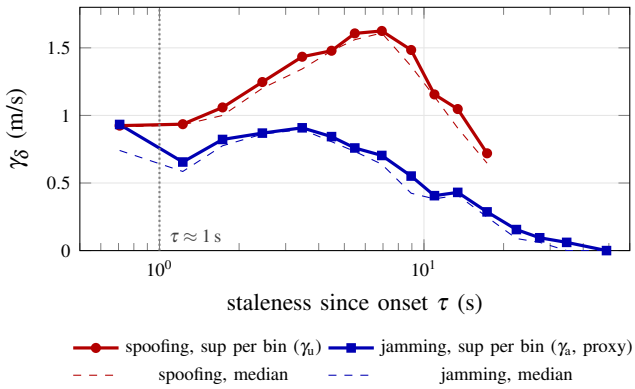


Fig. 2. The sensing side of the budget, measured on the real PX4 flights (EKF signal, baseline-corrected rate γ_δ of Eq. (3)). In the first second of staleness the spoofed and the jammed estimator degrade at the same rate: the spoofer has not yet walked the fix away from where an honest estimator would put it. Beyond that, the spoofer out-runs the jammed (degraded but honest) estimator by up to 1.74×. Authentication can buy at most the gap between the curves. The jamming flight is a proxy for rejection, because its receiver never lost the fix.

B. RQ2: What Authentication Costs

Table IV decomposes Δ_{auth} on classic CAN. Three observations stand out.

Latency is not the binding cost. Every per-message signature scheme that fits on the bus authenticates a message in at most 119.0 ms (certified), against the hundreds of milliseconds of budget computed in Section V-C. TESLA is the exception by design: its delay is the disclosure interval.

Occupancy is. ML-DSA-44 needs 346 frames per signature, so classic CAN admits only 16.4 signed messages per second (18.1 on an idle bus). One 10 Hz GNSS stream fits; two do not. Fig. 3 shows the cliff, and on CAN FD the admissible rate rises to 120.5/s. The DES of 254,992 transfers stays within the analytical bound in all configurations (a self-consistency check, as noted in Section IV), with a maximum of 2.56 ms against a bound of 5.44 ms for Ed25519, and 62.24 ms against 63.04 ms for ML-DSA-44. The bound is tight where it matters: for the long transfers that define the cliff. If navigation messages are released sporadically rather than periodically, the cliff softens into a slope (dashed in Fig. 3). The mean transmission latency alone then reaches the benefit threshold $C(0.25) = 371$ ms at about $0.91\,r^\star$, so the usable rate is practically the same.

SLH-DSA fails on the signer, not the verifier. At 480 MHz, an SLH-DSA-128s signature takes 15953.25 ms, about 16 s, while verification takes 15.566 ms. Its 7.9 kB signature also exceeds the bus at the required rate. Stateless hash-based signatures are usable on a flight controller only for rare artefacts, such as key-chain commitments or firmware, and not for navigation data.

C. RQ3: Where the Crossover Lies

Supremum discipline: At the operating point, the benefit threshold of Eq. (7) is $C(0.25) = 371$ ms (headline), 160 ms (conservative) and 816 ms (secant). Fig. 4(a) places the

TABLE IV

THE COST SIDE ON CLASSIC CAN (DRONECAN, 1 MBIT/S, $U_0 = 0.095$ DERIVED), AT 10 SIGNED MESSAGES PER SECOND. COMPUTE TIMES ARE LITERATURE CYCLE COUNTS SCALED TO 480 MHZ (LITERATURE); R_{tx} IS THE BUSY-PERIOD BOUND (MODEL). "CERT." USES THE $1 - 10^{-6}$ SIGNING TAIL FOR ML-DSA. "SRC" INDICATES SOURCE AUTHENTICATION AGAINST AN INSIDER; "PQ" INDICATES POST-QUANTUM SECURITY.

Scheme	tag (B)	frames n	t_{sign} (ms)	t_{ver} (ms)	R_{tx} (ms)	Δ_{auth} (ms)	cert. (ms)	$r^\star$ (msg/s)	Src	PQ
Group MAC (MAVLink 2 style)	13	3	0.06	0.057	4.3	4.4	4.4	1886.3	–	✓
Ed25519	64	10	1.03	2.636	5.4	9.1	9.1	565.9	✓	–
FN-DSA-512	666	96	46.81	0.827	19.2	66.8	66.8	59.0	✓	✓
ML-DSA-44	2420	346	4.43	1.694	63.0	69.2	119.0	16.4	✓	✓
SLH-DSA-128f	17088	2442	767.87	45.674	∞	∞	∞	2.3	✓	✓
SLH-DSA-128s	7856	1123	15953.25	15.566	∞	∞	∞	5.0	✓	✓
TESLA, 0.1 s interval	26	4	0.06	0.057	4.5	114.6	114.6	1414.7	✓	✓

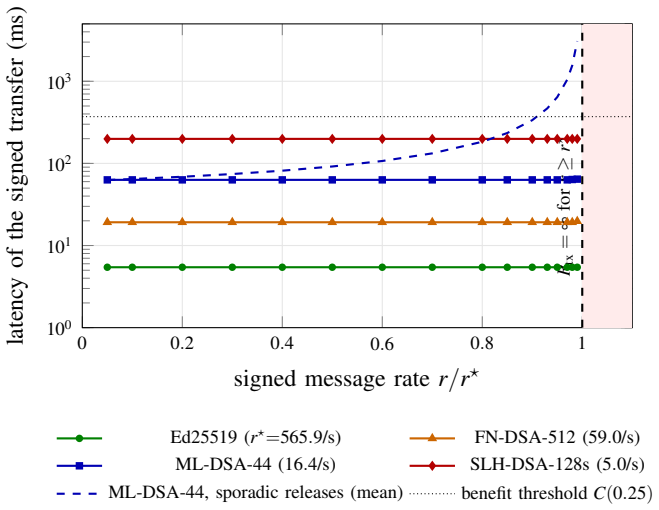


Fig. 3. The cost is a cliff, not a slope (Proposition 3). Solid lines show the busy-period bound of a signed transfer with periodic releases on classic CAN at 1 Mbit/s, above the base load $U_0 = 0.095$ derived from the HCRL UAVCAN capture. Below $r^\star$ the latency is essentially the transmission time of the signature; at $r^\star$ the busy period no longer closes, and the latency, and with it the certificate, is unbounded. With sporadic (Poisson) releases the cliff softens into a slope (dashed), which crosses the benefit threshold $C(0.25) = 371$ ms at $r \approx 0.91\, r^\star$.

schemes against $C(\theta)$. Under the headline ratio, the crossover setting is $\theta_\times = 0.080$ s for ML-DSA-44, 0.006 s for Ed25519 and 0.003 s for the group MAC. All lie below the benign floor, so every per-message scheme pays across the whole certified window. TESLA with intervals of 0.5 s or more does not, because its delay exceeds $C(0.25)$. Under the conservative ratio, ML-DSA-44 crosses at $\theta_\times = 0.186$ s, just above the benign floor of 0.178 s, so at the tightest admissible detector it does not pay.

Authentication rescues the tightest corridor: At $\theta = 0.25$ s the unauthenticated tube is 2.65 m, which does not fit $m = 2$ m. With authentication it shrinks to 1.53 m (group MAC), 1.55 m (Ed25519) and 1.88 m (ML-DSA-44), so all three certify the 2 m corridor under the headline scenario, conditional on the proxy for γ_a. Under the conservative scenario, ML-DSA-44 gives 2.48 m and the rescue fails. Of the 128 headline delay-regime cells of the selection grid, authentication is what makes 24 of them feasible.

TABLE V

HORIZON-MATCHED CROSSOVER (THEOREM 1) ON CLASSIC CAN. BENEFIT IS THE CHANGE IN TUBE RADIUS, $(\delta_u(\Delta) - \delta_a(\Delta + \Delta_{\text{auth}}))E$; A NEGATIVE VALUE MEANS THAT AUTHENTICATION ENLARGES THE TUBE. ENVELOPES MEASURED; COSTS AS IN TABLE IV.

Scheme	cert. Δ_{auth} (ms)	$\theta_\times$ (s)	gain at $\theta =$ 0.25 s (cm)	gain at $\theta =$ 0.5 s (m)
Group MAC	4.4	0.259	-3.4	1.11
Ed25519	9.1	0.260	-3.8	1.10
FN-DSA-512	66.8	0.274	-9.1	1.00
ML-DSA-44	119.0	0.284	-13.7	0.91
TESLA 0.1 s	114.6	0.283	-13.3	0.92
TESLA 0.5 s	514.6	0.999	-44.1	-0.48

Horizon-matched: Table V and Fig. 4(b) apply Theorem 1. At $\Delta(0.25) = 0.5$ s, both estimators have accrued about 0.46 m. The envelopes differ by 1 cm, far inside the 0.16 m residual noise of the error signal, so the data cannot show that authentication helps at the operating point. Taken at face value, the envelopes say that authentication enlarges the tube by 3.4 to 13.7 cm for the per-message schemes. The crossover moves to $\theta_\times = 0.259$–0.284 s. At $\theta = 0.5$ s the spoofed estimator has accrued 0.93 m against 0.65 m for the proxy, and authentication saves about one metre of tube.

TESLA illustrates the protocol term. With a 0.5 s interval it pays only from $\theta_\times = 0.999$ s; with a 1 s interval, only from 1.357 s.

Reconciling the two readings: The two readings answer different questions. The supremum discipline certifies a configuration for any staleness up to the window edge. It is the right object for an approval case, and under it, with the headline rates, authentication is never harmful for any per-message signature scheme that fits on the bus. The horizon-matched reading asks what the detector setting actually lets the adversary do. It is an empirical estimate, not a bound, because each envelope rests on one onset, but it is the right question for a design decision. On the measured spoofer, a tight detector already denies the spoofer the time it needs, so cryptography has nothing measurable left to buy. The location of the horizon-matched crossover belongs to one spoofer on one hover flight. A faster spoofer, still inside the EKF's innovation gate, moves it to the left. Experiment E3 maps that dependence.

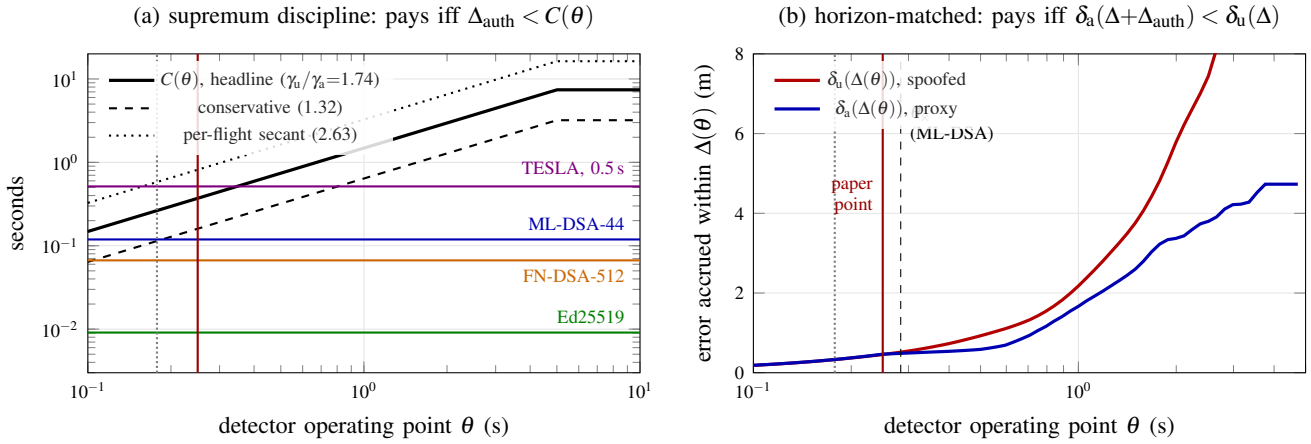


Fig. 4. The crossover map. (a) Under the supremum discipline of the parent certificate, one rate per configuration, the benefit threshold $C(\theta)$ grows linearly in θ until the slack saturates the hidden delay. Every per-message signature scheme that fits on the bus lies below the headline curve across the whole certified window. The vertical lines mark the benign floor (0.178 s) and the paper operating point (0.25 s). (b) Matching the horizon to the delay the detector actually admits changes the picture at tight settings. Within $\Delta(0.25) = 0.5$ s, the spoofed and the proxy estimator have accrued the same error to within 1 cm, so the data cannot show a benefit. On these envelopes, authentication pays from $\theta_\times = 0.284$ s for ML-DSA-44 (Table V). Both panels use the jamming flight as a proxy for γ_a.

TABLE VI
DOES FRESHNESS BOUND THE HIDDEN DELAY? BINDING REQUIRES $\tau_{\text{fresh}} + \varepsilon_{\text{clk}} < \Delta(\theta)$. THE HOLDOVER COLUMN GIVES HOW LONG A FREE-RUNNING XO (±20 PPM) KEEPS A TIGHT WINDOW BINDING AT $\theta = 0.25$ S. SPECIFICATIONS LITERATURE; DRIFT MODEL.

Window	τ_{fresh} (s)	binds?	note
MAVLink 2 signing	60	no	6× the largest Δ
OSNMA T_L (ADKD0/4)	30	no	meaconing below T_L passes
OSNMA receiver maximum	165	no	
tight, $\tau_{\text{fresh}} = \theta$	0.25	yes	XO holdover ≈3.5 h

D. RQ4: Freshness and the Clock

Table VI evaluates Proposition 5.

No deployed window binds. The largest delay the detector admits, $Hs = 10$ s, is 3–17 times smaller than the windows in use, so in the delay regime the detector, not the signature, bounds the hidden delay. A window tight enough to matter must be set at detector scale. Natural drift is then harmless: a ±20 ppm crystal accumulates 72 ms per hour, so a 0.25 s margin survives about 3.5 h of GNSS-free holdover.

The danger lies in adversarial time. Under a forward-only update from GNSS time (Proposition 5(b)), a spoofer advancing GPS time at 100 ms/s (or 10 ms/s) makes all legitimate signed traffic stale after 600 s (or 6000 s); a step change does so immediately. Even a slow push of 1 ms/s needs only 16.7 h, although at 1000 ppm such a push is far outside any oscillator's error and a rate-plausibility check would catch it. The result is a persistent-denial adversary manufactured from authentication itself.

E. RQ5: Induced Verification

Table VII reports ρ_{iv} and the bus share a_{sat} that saturates a verifier with $\beta = 0.2$.

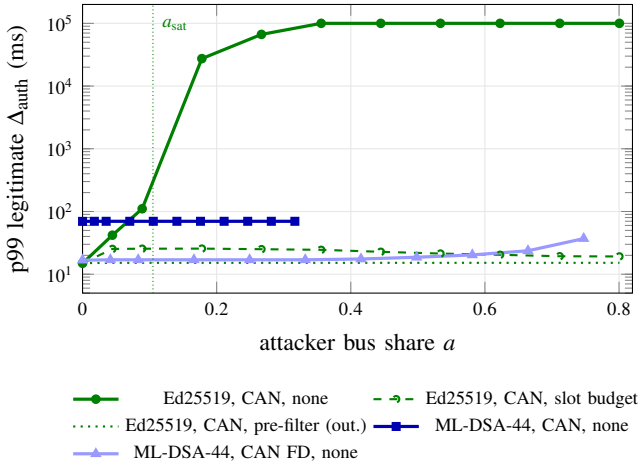


Fig. 5. Induced verification (Proposition 6), discrete-event simulation with verification CPU share $\beta = 0.2$ and the HCRL base load. Forged Ed25519 transfers cost the verifier 1.647 CPU-seconds per bus-second (ρ_{iv}). Beyond $a_{\text{sat}} = 0.105$ of the bus, the verification queue diverges and the legitimate signed stream is denied (values above 10^5 ms are clipped). ML-DSA-44 on classic CAN is immune, because its signature is too expensive to *transmit* for a flood to reach the CPU. On CAN FD the same scheme moves to $\rho_{\text{iv}} = 0.226$.

Two results invert common intuition. First, the classical scheme is the most exposed: an Ed25519 signature is cheap to send and comparatively expensive to verify. Second, post-quantum migration on classic CAN *removes* the exposure, because ML-DSA-44 and FN-DSA-512 cost more bus time than verification time. A faster bus restores it: on CAN FD, ML-DSA-44 has $\rho_{\text{iv}} = 0.226$, and a flood of about 80 % of the bus saturates the verifier.

The pre-filter removes the outsider's leverage entirely but leaves the insider's unchanged, as Table I predicts. The slot budget bounds the induced delay to one verification for both,

TABLE VII

VERIFICATION AMPLIFICATION $\rho_{\text{iv}} = t_{\text{ver}}/(nC_{\text{f}})$, THE ATTACKER BUS SHARE a_{sat} THAT SATURATES A VERIFIER AT $\beta = 0.2$, AND THE LARGEST SHARE $a_{\max} = 1 - U_0 - rnC_{\text{f}}$ THE ATTACKER CAN OBTAIN. THE VERIFIER IS EXPOSED IFF $a_{\text{sat}} < a_{\max}$. MODEL ON LITERATURE COSTS.

	classic CAN			CAN FD		
Scheme	ρ_{iv}	a_{sat}	$a_{\max}$	ρ_{iv}	a_{sat}	$a_{\max}$
Group MAC	0.119	1.681	0.901	0.296	0.674	0.903
Ed25519	1.647	0.105	0.889	6.842	0.025	0.902
FN-DSA-512	0.054	3.561	0.752	0.390	0.491	0.884
ML-DSA-44	0.031	5.981	0.352	0.226	0.811	0.830
SLH-DSA-128s	0.087	–	n/a	0.647	0.069	0.665

TABLE VIII

SELECTION RULE OUTPUT (ALGORITHM 1), HEADLINE γ, DELAY REGIME, INSIDER ADVERSARY, POST-QUANTUM REQUIRED, CLASSIC CAN. "NONE" MEANS THAT THE UNAUTHENTICATED CERTIFICATE ALREADY HOLDS; ML-DSA DENOTES ML-DSA-44.

θ (s)	$m = 2$ m	$m = 5$ m	$m = 10$ m	$m = 20$ m
0.178	none	none	none	none
0.25	ML-DSA, k=1	none	none	none
0.5	infeasible	ML-DSA, k=4	none	none
1.0	infeasible	infeasible	ML-DSA, k=8	none

and it is the only mitigation that makes Ed25519 flood-safe against an insider.

F. Amortisation and Selection

Batching moves a scheme along the trade-off of Eqs. (12)–(13). For ML-DSA-44 on classic CAN, $k = 8$ lowers bus utilisation from 64.8 % to 17.7 %, at a certified latency of 820 ms. That latency is affordable only on wide corridors. Table VIII shows the rule's choices for the most demanding requirement: an insider adversary with post-quantum security mandated.

The pattern generalises across the grid. Authentication matters exactly on the boundary of the unauthenticated certified window, where it buys one corridor size. The rule spends any slack beyond that on batching, to save bus capacity. In the denial regime, where no unauthenticated configuration is certified (Proposition 4), the rule finds a certified configuration in all but 24 of 128 headline cells.

VI. DISCUSSION

A sound certificate is not a design oracle: The supremum and horizon-matched readings agree on everything except the decision at tight detector settings, and there they disagree completely. We draw a general lesson. Certificates built to be conservative inherit their conservatism unevenly across the alternatives they compare. A design choice should be read from the bound that is tight for both alternatives, not from the one that is safe for each.

Guidance for post-quantum migration: For navigation data on a classic CAN flight-controller bus, ML-DSA-44 at one 10 Hz stream is admissible, certified, and flood-immune. FN-DSA-512 is limited by its signing cost on the MCU unless the signer is a companion computer. SLH-DSA belongs off the navigation path. Moving to CAN FD relieves occupancy but reopens the induced-verification attack, so a slot budget should accompany the migration. Against an insider, group MACs, including MAVLink 2 signing, must be supplemented by origin authentication.

Time is part of the trusted computing base: Freshness that matters must be set at detector scale. It must then be anchored to a clock the GNSS adversary cannot move. Network Time Security [34] over a two-way link is one option. Bounding the rate at which GNSS time may update the signing clock is another, and the cheapest.

Regulatory reading: The budget turns a security configuration into a quantity that an operational approval already uses: the lateral extent of the SORA contingency volume [14]. For an assurance case under the airworthiness security process specification DO-326A [35], the selection rule's output is a traceable argument that a given authentication configuration keeps a declared corridor.

VII. RELATED WORK

Table IX positions the paper against the closest work.

Two observations sharpen the gap. The only conversions of authentication latency into distance we found are kinematic: latency multiplied by vehicle speed, as when in-vehicle authentication delays such as vatiCAN's 3.3 ms [5] are related to distance travelled. Such a conversion is neither certified nor sensitive to which source the estimator trusts. Recent work on PQC transport over CAN FD recommends FN-DSA-512 on feasibility grounds [39]. Our analysis agrees on occupancy, but adds that FN-DSA's signing cost on a 480 MHz MCU exceeds a 20 % processor share at 10 Hz.

VIII. LIMITATIONS AND THREATS TO VALIDITY

The authenticated rate is a proxy. The jamming flight kept a degraded three-dimensional fix throughout, so γ_{a} measures an estimator fed honest but noisy GNSS, not one that has rejected GNSS. Every conclusion that compares γ_{u} and γ_{a} inherits this proxy, and E1 and E3 replace it with direct measurements of dead reckoning after rejection.

The sensing side is one airframe in hover. Both rates come from one spoofing and one jamming flight of a PX4 Holybro S500 [11], inherited from the parent. The spoofer's walk rate was chosen by the original experimenters; it is not a worst case. Platform, flight mode and airspeed are not varied. E3 is designed to vary them in simulation.

Compute costs are scaled literature values. Cortex-M4 and M7 cycle counts measured at 24 MHz will change at 480 MHz with flash wait states and caches. ML-DSA's tail is modelled, not measured, and FN-DSA's tail is not modelled. E4 replaces all of these.

The transport model is worst-case and partly assumed. The HCRL payload mix is measured, but the bitrate (1 Mbit/s) is assumed: the capture's timestamps are too coarse to check it, and the DroneCAN framing rules should be confirmed against

TABLE IX
POSITIONING AGAINST PRIOR WORK.

Line of work	What it reports	What this paper adds
CAN authentication: vatiCAN [5], LeiA [6], CANAuth [36], S2-CAN [7], CANsec [15]	latency in ms, bus-load increase, memory	the same costs priced in metres of certified tube; the insider gap of group keys formalised as an adversary class
PQC on Cortex-M: pqm4 [24], SLOTHY [25], Howe–Westerbaan [37], LMS/XMSS [38]	cycles, stack	signer-side infeasibility and the occupancy cliff $r^\star$; tail-certified signing time
PQC transport over CAN FD [39]	feasibility under contention	closed-form $r^\star$, response-time analysis on a measured base load, and the reversal of flood exposure
GNSS authentication: OSNMA [1], [40], Chimera [2]	time to first authenticated fix, loose-synchronisation bound	T_L as a freshness window inside the certificate; the meaconing limit on γ_a
Time attacks on authentication [17], [18], [22], [23]	attacks and oscillator requirements	forward time push converting delay into denial, as a regime change of the certificate
Broadcast authentication and DoS resistance [31], [41]–[44]	disclosure delay, pre-filters	the amplification ratio ρ_{iv} and its scheme- and bus-dependent reversal
Security-aware real-time design [45]–[47]	MAC allocation under deadlines, by utilisation	a selection constrained by a certified physical error bound
Certified robustness, tubes and runtime assurance [48]–[53]	robustness to a given perturbation	where the perturbation comes from, and what removing part of it costs
Parent composition [8]	γ and Δ as fixed inputs	both as design variables traded through authentication

the specification. The busy-period bound treats the background as periodic. Both need E5's hardware arm.

Additivity. Eq. (5) adds Δ_{auth} to the hidden delay. This is conservative when the authenticated message is not on the delayed path.

Single-onset envelopes. The horizon-matched result at the operating point rests on a 1 cm difference between two single-onset envelopes. It shows that the data cannot distinguish the two configurations there, not that authentication is harmful.

External validity. The crossover inequality and the cliff are structural; their locations are platform-specific. We therefore release the calculator, not only the numbers.

IX. Conclusion

Authentication is neither free nor automatically beneficial for a UAV's navigation. Inside a certified mission-completion budget, it buys a lower error rate and pays in delay, and the exchange rate can be measured. On the data available today: the cost on a flight-controller bus is a cliff in occupancy, not a slope in latency; the benefit appears only once a spoofer has had time to out-run an honest estimator; freshness windows in use are too loose to bound delay, and the clock that would tighten them can be turned against them; and the scheme most exposed to forged-signature floods is the classical one. The certified selection rule turns these findings into a configuration choice, and the released artifact turns each remaining modelled input into a defined experiment.

Ethics Considerations

No human subjects or personal data are involved. All flight and bus data are public research corpora, used under their licences and not redistributed. The time-push and induced-verification analyses describe attacker capabilities. Each is presented with its mitigation: bounding the GNSS-to-clock update rate, and slot budgeting. The time-push behaviour follows from the public MAVLink specification and has been reported experimentally in prior work [22]. Before publication we will notify the MAVLink maintainers of the conversion from delay to denial and of the rate-bound mitigation.

Open Science

The calculator, simulations, result files, figure data and the runbook for the remaining experiments are released in the anonymised artifact. Every number in this paper is a macro generated from those files.

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APPENDIX A
PROOFS

Proof of Proposition 1. Authentication reduces the error if and only if $(\Delta+\Delta_{\text{auth}})\gamma_{\text{a}}E < \Delta\gamma_{\text{u}}E$. Dividing by $\gamma_{\text{a}}E > 0$ gives Eq. (7). For $\theta < s$, $\Delta = H\theta$, and solving for θ gives $\theta_{\times}$. □

Proof of Proposition 2. On $\theta < s$, $\Delta(\theta) = H\theta$, so $dB/d\theta = -H$ and $dC/d\theta = H(\gamma_{\text{u}}/\gamma_{\text{a}} - 1) > 0$. □

Proof of Theorem 1. Replace the interface mapping $\delta = \gamma\Delta$ of the parent with the envelope $\delta(\Delta)$. This remains an upper bound on the perturbation for any onset in the observed set, because δ is a supremum over onsets and over staleness up to Δ. The Grönwall step is unchanged, so the tube radii are $\delta_{\mathrm{u}}(\Delta)E$ and $\delta_{\mathrm{a}}(\Delta+\Delta_{\mathrm{auth}})E$; comparing them gives Eq. (8). □

Proof of Proposition 3. With periodic releases on a non-preemptive fixed-priority bus, the level-i busy period of the signed stream is the least fixed point of $t = \lceil tr \rceil nC_{\mathrm{f}} + \sum_j \lceil t/T_j \rceil C_j$. A fixed point exists if and only if the total utilisation $rnC_{\mathrm{f}} + U_0 < 1$ [21], that is, if and only if $r < r^{\star}$. Within a finite busy period, the response time of each instance is the finite fixed point of its completion-time recurrence. Without one, the backlog grows without bound. □

Proof of Proposition 4. Under $\mathscr{A}_{\mathrm{deny}}$, no uncorrupted update arrives within T_c, so the re-anchoring hypothesis of the parent fails and its tube grows with the denial duration. An authenticated source cannot be forged, so either it is delivered, and re-anchoring holds, or it is withheld, which is the denial case of that source alone. □

Proof of Proposition 5. (a) The hidden delay is $\min(\Delta, \tau_{\mathrm{fresh}} + \varepsilon_{\mathrm{clk}})$, which is smaller than Δ only if $\tau_{\mathrm{fresh}} + \varepsilon_{\mathrm{clk}} < \Delta$. (b) Under the update rule, the verifier's clock after pushing for time t is at least the true time plus νt, while a legitimate sender's timestamp is the true time. A packet is rejected when it is more than τ_{fresh} behind, which happens for $t > \tau_{\mathrm{fresh}}/\nu$. The forward-only update prevents recovery until the adversary stops and real time catches up. □

Proof of Proposition 6. Verification is a single-server queue with offered load $(a/(nC_{\mathrm{f}}))t_{\mathrm{ver}} + rt_{\mathrm{ver}}$ per unit time, served at rate β; it is stable if and only if the load is below β. Under the slot budget, at most $k_s - 1$ forged candidates precede the legitimate one in any slot, so the legitimate message waits for at most $(k_s - 1)t_{\mathrm{ver}}/\beta$ beyond its own service. □

Appendix B
Acronyms

CAN / CAN FD	controller area network / with flexible data rate
CANsec	CAN XL security layer
DES	discrete-event simulation
EKF	extended Kalman filter
FN-DSA	FFT-over-NTRU-lattice digital signature algorithm (Falcon)
GNSS / GPS	global navigation satellite system / global positioning system
IMU	inertial measurement unit
JARUS SORA	Joint Authorities for Rulemaking of Unmanned Systems, Specific Operations Risk Assessment
MAC	message authentication code
MAVLink	Micro Air Vehicle Link
MCU	microcontroller unit
ML-DSA	module-lattice digital signature algorithm
OSNMA	Open Service Navigation Message Authentication
PQ / PQC	post-quantum / post-quantum cryptography
RQ	research question
SITL	software in the loop
SLH-DSA	stateless hash-based digital signature algorithm
TESLA	Timed Efficient Stream Loss-tolerant Authentication
UAV	unmanned aerial vehicle
VIO	visual–inertial odometry
XO / TCXO / OCXO	crystal / temperature-compensated / oven-controlled oscillator